

U03-02.TodoAPI-ImplementAPI.md

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RECORD OF CHANGES

[illegible]

Exercise: FastAPI + Async SQLAlchemy + Basic TODO CRUD

Goal

Build an API-only TODO service using **FastAPI + SQLAlchemy 2.x async** (asyncpg) connected to the Dockerized Postgres.

Suggested Project Structure

```
app/  
  core/config.py  
  db/session.py  
  db/models.py  
  db/repository.py  
  api/routes/todos.py  
  main.py  
alembic/ (created in Lab 4)  
compose.yaml (later adds app service)  
Dockerfile
```

Tasks

1. Dependencies

```
pip install fastapi uvicorn sqlalchemy asyncpg pydantic pydantic-settings
```

2. Async SQLAlchemy setup

- Depend on your implementation

3. Model

- Todo model has some fields: id, title, description, completed, created_at, updated_at

4. Schema (Pydantic)

- Create schema models: TodoCreate, TodoRead with require validation

5. FastAPI App

- Consider to use router (APIRouter - fastapi)

6. Run

```
uvicorn app.main:app --reload  
# Test in /docs (Swagger) and with curl  
curl -X POST http://localhost:8000/todos -H "Content-Type: application/json" -d '{"title":"Read paper"}
```

Acceptance Criteria

- API responds with 201 on create, returns JSON aligned with schema.
- GET, PATCH toggle, DELETE behave as defined.
- /docs loads correctly with generated OpenAPI.

Deliverables

- Source code.
- A short README with run instructions and sample curl or test script calls.

Rubric (40 pts)

- Async DB setup
 - Model + schema correctness
 - CRUD endpoints working
 - README & API docs
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Internal